

A model for monazite/melt equilibrium and application to the generation of granitic magmas

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ABSTRACT

In order to model the behaviour of Zr, Hf, Ti, P, U, Th, Y and the rare-earth elements (REE) in the continental crust, the parameters which control the equilibria between accessory minerals and granitic melts must be determined. This paper proposes equations for the equilibrium between monazite and Ca-poor felsic melts. In a first step, all the light REE are considered as a single component and an equation describing monazite solubility is deduced from the available experimental data. In a second step, the fractionation of REE by monazite is investigated on the basis of a natural example of coexisting monazite and obsidian from Macusani, Peru. The solubility of monazite can be used as a thermometer, based on the total REE content of natural magmatic rocks. Application of REE thermometry to various peraluminous rock series suggests that it is a reliable tool for determining temperatures of magmatic processes. The equation governing REE fractionation by monazite is used to discuss genetic relationships between a melt and a presumed parental rock, using the composition of the monazite in the parental rocks and the REE pattern of the daughter rock. This method has been applied successfully to High Himalayan Manaslu and Tibetan Slab granites. It is suggested that detailed petrographical work on monazite including electron microprobe analyses and SEM imaging of its internal structures, combined with the model presented here, can be a useful tool for deciphering the genesis and the differentiation of peraluminous granitic magmas in the continental crust.

1. Introduction

In contrast with mafic or intermediate suites, for which the evolution of trace-element concentrations is governed by relatively simple laws involving the major minerals (Allègre and Minster, 1978), it is now well established that accessory minerals play the dominant role in felsic suites. Zircon, apatite, monazite, sphene, allanite and xenotime, for example, control the behaviour of trace elements such as Zr, Hf, Ti, P, U, Th, Y and the rare-earth elements (REE). As a consequence, the evolution of most isotopic systems, and most of the commonly used geochronometers (Sm/Nd, U/Pb, Th/Pb, Lu/Hf) are dominated by accessories. A comprehensive knowledge of the behaviour of these minerals during the main magmatic processes is therefore required in order to interpret cor-

rectly the isotopic characteristics and ages of felsic rocks. Such data will become even more important as *in situ* isotopic analyses [secondary ion mass spectrometry (SIMS), laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS)] become more routine and their application more widespread in geological laboratories.

The ubiquitous distribution of monazite in high-grade metasediments and in peraluminous granites suggests that it controls the behaviour of REE, U and Th during the formation and differentiation of these magmas (Overstreet, 1967; Miller and Mittlefehldt, 1982; Montel, 1986; Rapp and Watson, 1986; Cuney and Friedrich, 1987). It is supported by the available experimental data (Montel, 1986; Rapp and Watson, 1986) which demonstrate that for the range of temperature and melt

composition involved during partial melting of metasediments the behaviour of light REE (LREE) should be governed by chemical equilibria between monazite and melt.

The aims of this paper are to propose: (1) an equation describing the solubility of monazite in Ca-poor felsic melt based on the available experimental data; and (2) a set of equations that permits calculation of the REE pattern of a melt in equilibrium with monazite. Two types of application of these data will be given: a thermometer based on REE abundance in granites (REE thermometry), and a method for discussing genetic relationships between a source rock and its presumed product, based upon monazite/melt equilibria.

2. General equation for monazite solubility

2.1. Experimental data

Two sets of experimental data are available for the solubility of monazite in felsic melts. Rapp and Watson (1986) studied the solubility of a natural monazite in a natural, Ca-, Fe- and Mg-poor, obsidian glass. Experiments were carried out at 8 kbar, in the range 1000–1400°C, 1–6 wt% H₂O (i.e. one melt composition, *T* and H₂O being variable). These were partial dissolution experiments, so the solubility of the phase and the diffusion parameters of the elements are obtained from the gradient of concentration in the glass around a large crystal of monazite. Their results indicate a strong positive dependence of monazite solubility on temperature (Arrhenius-type relationship), and a more limited effect of the water content. Montel (1986) employed a different experimental strategy, using a synthetic system consisting of Ce-monazite (CePO₄) and Na₂O–K₂O–SiO₂–Al₂O₃ melts. Runs were carried out at 2 kbar, 800°C, under water-saturated conditions (i.e. *T* constant, melt composition variable). His results indicate a strong effect of the melt composition, monazite being much more soluble in peralkaline melts than

in metaluminous melts. Additional experiments in the synthetic system at 900° and 1000°C (Montel, 1987) confirmed the effect of *T*. Rapp et al. (1987) confirmed the influence of the melt composition by determining the solubility of monazite in some natural Fe-, Mg- and Ca-rich melts. Ellison and Hess (1988) found that LaPO₄ is more soluble in peraluminous melts, in the system SiO₂–Al₂O₃–CaO–K₂O at 1 atm, than in peralkaline melts. Montel (1986, this study), and Rapp et al. (1987) were unable to detect such an effect, but few experiments were performed with true peraluminous compositions.

In spite of very different experimental techniques, the data of Montel (1976, 1987, this study) and Rapp and Watson (1986) agree remarkably well (Fig. 1). This confirms the reliability of both sets of data, and suggests that pressure has little, if any, effect on monazite solubility, since the experiments were carried out at different pressure. It suggests also that monazite solubility can be simply described as the total amount of REE dissolved in the melt, whatever monazite is considered: pure CePO₄ or complex natural monazite.

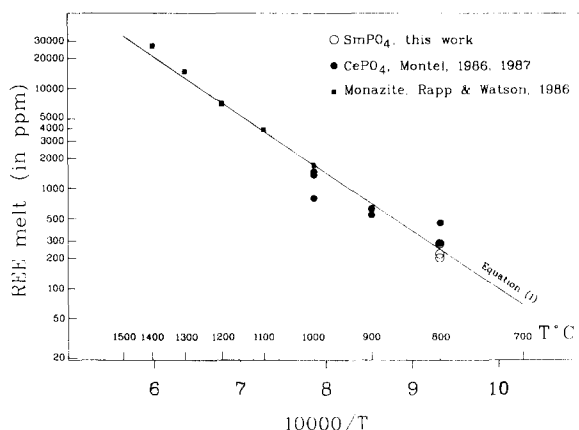


Fig. 1. Solubility of monazite in felsic melts, under water-rich conditions, as determined by Rapp and Watson (1986), for $D=1.08$, and Montel (1986, 1987, this work), for $D=1.03$ – 1.17 . Note the good agreement between the two sets of experimental data.

2.2. Resulting expression

A generalized expression has been derived by assuming that melt composition, temperature and water content independently affect monazite solubility. There are two justifications for this assumption: (1) reducing the water content of the melt simply shifts the Arrhenius-type relationship between the solubility of monazite and T toward lower values, with only a very small effect on the slope (Rapp and Watson, 1986, fig. 6). (2) Increasing the temperature increases the solubility of monazite but does not affect the slope of the relationship between D and the REE content of the melt (Montel, 1987).

The final expression was established by determining each coefficient independently using the relevant experiments. A final check was done to insure that the expression fit the experimental data properly. The calculated expression is:

$$\ln(\text{REE}_i) = 9.50 + 2.34D + 0.3879\sqrt{\text{H}_2\text{O}} - 13318/T \quad (1)$$

where

$$\text{REE}_i = \sum \frac{\text{REE}_i \text{ (ppm)}}{\text{at. weight (g mol}^{-1}\text{)}}$$

and

$$D = \frac{(\text{Na} + \text{K} + \text{Li} + 2\text{Ca})}{\text{Al}} \cdot \frac{1}{(\text{Al} + \text{Si})}$$

Na, K, Li, Ca, Al, Si are in at%; H_2O is in wt%; and T in K. The REE considered extend from La to Gd, excluding Eu.

This expression does not fit the data of Rapp et al. (1987) which are also in disagreement with the results of Rapp and Watson (1986). The compositions used by Rapp et al. (1987) are rich in Ca, Fe and Mg, and this can significantly change monazite solubility. In particular, Ca is known to destabilize monazite to form allanite (Lee and Bastron, 1967; Cuney and Friedrich, 1987). Because the effect of Ca,

Fe and Mg upon monazite solubility were not taken into account in Eq. 1, applicability of the expression is probably restricted to Ca-, Fe- and Mg-poor compositions.

2.3. Effect of the various parameters

Three parameters influence the solubility of monazite in expression (1): the temperature, the water content of the melt and the melt composition. Temperature is the most important parameter: for $D=1$ and $\text{H}_2\text{O}=6\%$, the solubility of monazite according to Eq. 1 is 27 ppm at 650°C , 57 ppm at 700°C , 204 ppm at 800°C and 589 ppm at 900°C . A small variation of T produces a large variation in the REE content of a melt coexisting with monazite. D is also an important parameter. At 800°C and $\text{H}_2\text{O}=6\%$, the solubility increases from 204 ppm for $D=1$ to 660 ppm for $D=1.5$, which corresponds, in natural rocks, to a highly alkaline composition. For peraluminous or metaluminous crustal melts D varies from 0.9 to 1.1; typical values are between 0.95 to 1.05. Li is important only for highly differentiated magmas. The water content plays only a limited role. Changing H_2O from 4% to 8% (i.e. doubling the water content) produces an increase in REE from 171 to 204 ppm for $D=1$ and $T=800^\circ\text{C}$, but the effect is important for low water contents. Rapp and Watson (1986) emphasized that the water content is important also for the rate of dissolution of monazite.

When recast to similar units, Eq. 1 yield values almost identical to those produced from the calibration of Rapp and Watson (1986; numerically formulated in Rapp et al., 1987), for D close to 1 and H_2O in the range 4–8%. Rapp et al. proposed that their calibration was valid for all peraluminous melts. Eq. 1 is valid also for the peralkaline composition, provided that the Ca, Fe and Mg contents are not too high.

2.4. Effect of monazite composition

Eq. 1 is designed only for a monazite close to pure REEPO_4 , such as those used in the ex-

perimental studies. In natural monazite the incorporation of Th- and U-bearing end-members [huttonite (ThSiO_4), brabantite ($\text{CaTh}(\text{PO}_4)_2$) and the uraniferous equivalents] may significantly lower the activity of REEPO_4 (Deer et al., 1962). Assuming that the Th and U end-members act as inert diluents, REE_i in Eq. 1 must be replaced by $\text{REE}_i/X_{\text{REEPO}_4}$, where X_{REEPO_4} is the sum of the mole fractions of REE-phosphates in the monazite. Compilation of 35 microprobe analyses carried out at Clermont-Ferrand on high-grade gneisses and peraluminous granite yields an average value of 0.83 for X_{REEPO_4} . If the composition of the monazite is unknown, it is probably better to use this value rather than taking $X_{\text{REEPO}_4} = 1$.

3. Fractionation of REE by monazite

3.1. Evidence for REE fractionation by monazite

In Eq. 1 all the REE are considered as a single component. This is a convenient but simplistic approximation. Important information is lost in working just with the sum of the REE, since the shape of a REE pattern is at least as informative as the total REE content. As noted initially by Miller and Mittlefehldt (1982), the REE depletion observed in granitic series is associated with a "flattening" of the chondrite-normalized REE pattern, because the La/Sm ratio decreases with increasing differentiation. If this type of REE evolution is due to the extraction of monazite, it implies that monazite is able to fractionate the REE by incorporating preferentially lighter REE relative to heavier ones. This point is clearly demonstrated in the Macusani (Peru) glasses. These peraluminous obsidian glasses, contain andalusite, sanidine, plagioclase, biotite and muscovite as microphenocrysts. Accessory monazite and zircon have been reported mainly as inclusions in andalusite (Pichavant et al., 1987a, 1988). A systematic investigation has shown that mon-

azite, zircon and xenotime are present as microphenocrysts (10–50 μm) in the glass. The compositions of the monazites have been determined by microprobe (Table 1, see analytical procedure in Section 5.1), and are compared to the composition of the coexisting glass (JV2; Pichavant et al., 1987a, table 4). Relative to the glass, monazite has definitely a strong preference for the LREE, showing a steep chondrite-normalized pattern in contrast with the relatively flat REE pattern of the glass (Fig. 2).

3.2. Origin of fractionation

Fractionation of REE by monazite can have two origins: (1) each REE-phosphate has a distinct solubility, or (2) all the REE-phosphates have similar solubilities and fractionation results from non-ideality in the monazite solid solution.

In order to choose between these two possibilities, the solubility of pure SmPO_4 has been determined experimentally at 800°C, 2 kbar. The experimental and analytical procedures are identical to those used by Montel (1986) except for: (1) only dissolution runs were done; (2) an automated Caméca® SX-50 electron microprobe was used to analyse the run products, with synthetic glasses analysed by isotope dilution as standards for Sm; and (3) an internally heated vessel, working horizontally, pressurized with Ar was used. Results are reported in Table 2, and are compared to the solubility of CePO_4 in Fig. 3. CePO_4 and SmPO_4 have similar solubilities at 800°C, for any value of D . Therefore, it can be reasonably assumed that all the LREE-phosphates (La to Sm) have similar solubilities, and that Eq. 1 is valid for any REE-phosphate with a monazite structure. This explains the good agreement between the data obtained by Rapp and Watson (1986) using a natural monazite, and the solubilities determined for CePO_4 and SmPO_4 . It is also in agreement with the results of Ellison and Hess (1989) who showed that the partition coeffi-

TABLE 1

Microprobe analyses and structural formula of monazite from the Macusani glass

Crystal No.	1	2	6	11	10	13
ThO ₂	7.21	7.24	7.54	7.28	7.38	7.03
P ₂ O ₅	30.48	31.35	29.73	30.30	29.29	30.05
Y ₂ O ₃	1.93	2.28	2.33	2.33	2.45	2.46
CaO	1.83	1.95	2.79	2.03	1.89	1.96
UO ₂	1.65	2.11	2.81	2.32	2.21	2.50
Gd ₂ O ₃	1.67	1.71	2.13	1.73	1.51	1.61
Sm ₂ O ₃	2.30	2.24	3.01	2.40	1.96	2.17
Nd ₂ O ₃	11.64	11.00	11.18	11.11	10.44	10.81
Pr ₂ O ₃	3.03	3.05	2.90	2.93	2.67	2.71
Ce ₂ O ₃	25.75	25.80	23.05	23.98	23.59	23.66
La ₂ O ₃	11.98	11.98	9.79	10.86	11.12	10.91
SiO ₂	1.01	0.28	2.29	0.42	5.15	1.31
Total	100.49	100.99	99.55	97.71	99.66	97.19
Si	0.038	0.010	0.083	0.016	0.172	0.049
P	0.962	0.990	0.917	0.984	0.828	0.951
La	0.165	0.165	0.132	0.154	0.137	0.150
Ca	0.351	0.352	0.307	0.337	0.288	0.324
Pr	0.041	0.041	0.038	0.041	0.032	0.037
Nd	0.155	0.146	0.145	0.152	0.125	0.144
Sm	0.030	0.029	0.038	0.032	0.023	0.028
Gd	0.021	0.022	0.027	0.023	0.017	0.021
Y	0.038	0.045	0.045	0.048	0.044	0.049
U	0.014	0.018	0.023	0.020	0.016	0.021
Th	0.061	0.061	0.062	0.063	0.056	0.060
Ca	0.073	0.078	0.109	0.084	0.068	0.078
(La/Gd) _N	5.71	5.57	5.76	3.65	5.85	5.39

Monazites 1 and 2 occur as inclusions in andalusite (Pichavant et al., 1988; fig. 6c). Others are in the glass. All crystals are apparently homogeneous. Structural formula calculated assuming Si+P=1. (La/Gd)_N: La/Nd normalized to chondrites.

cients between two immiscible silicate liquids are identical for any REE, and that differences in crystal/melt partitioning are due to the properties of the solid and not of the melt (Ellison and Hess, 1989, p.1973).

3.3. A model for monazite solid solution properties

As indicated above (Section 3.2), fractionation of the REE is likely to result from non-ideality in the monazite solid solution. No experimental data are available relevant to this

problem, but the fractionation measured the Macusani glass–monazite pairs have been used to build a simple model for monazite/melt equilibrium. Only REE-phosphates are taken into account, Th and U end-members being considered as inert diluents. For each REE-phosphate, REE, in Eq. 1 is replaced by:

$$\text{REE}_i \rightleftharpoons \frac{\text{REE}_i}{X_{\text{REE}_i\text{PO}_4} \cdot \gamma_i} \quad (2)$$

where

$$\text{REE}_i = \frac{\text{REE}_i \text{ (ppm)}}{\text{at. weight (g mol}^{-1}\text{)}}$$

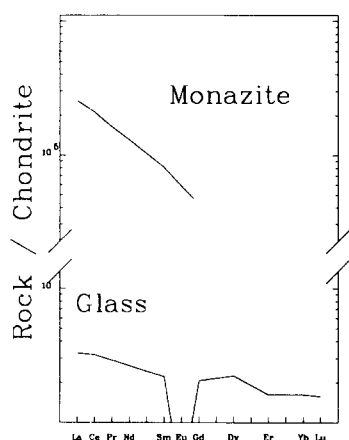


Fig. 2. Chondrite-normalized pattern for the Macusani glass, (JV2; Pichavant et al., 1987a, table 4) and the co-existing monazite (this study, Table 1, average composition). Chondrite values from Evensen et al. (1978), multiplied by 1.5.

X_{REEiPO_4} = mole fraction of REE-phosphates in monazite, given by the structural formula (see Section 5.1); and γ_i = activity coefficient such as defined by:

$$RT \ln \gamma_i = W_i \quad (3)$$

in which T is the absolute temperature in K; and W_i is a constant.

In this formulation the activity coefficients are temperature-dependent, assuming that

monazite solid solution is more ideal as temperature increases. W_i can be estimated from the composition of monazites (Table 1, average composition) and of the glass (JV2; Pichavant et al., 1987a, table 4). γ_i is calculated from:

$$\gamma_i = \frac{\text{REE}_i}{\text{REE}_t \cdot X_{\text{REEiPO}_4}} \quad (4)$$

Assuming a temperature of 590°C for the Macusani glasses (see discussion below), Eq. 4 yields the values for W_i/R given in Table 3. Combining Eqs. 1–3 it is possible to calculate the REE pattern of a melt in equilibrium with a monazite of known composition, or the composition of a monazite crystallizing from a melt of known composition.

The values for W_i obviously depend on the estimated temperature of monazite/melt equilibrium in the Macusani glass. The temperature taken is, for internal consistency, that given by Eq. 1 which is slightly lower than the temperature estimated for the main magmatic stage (650°C; Pichavant et al., 1988, p.318), but it is not an unrealistic temperature compared with the experimental results of Pichavant et al. (1987b) and London et al. (1989).

TABLE 2

Solubility of SmPO_4 at 800°C, 2 kbar, under water-saturated conditions, in SiO_2 – Al_2O_3 – Na_2O – K_2O melts

Sample No.	S18-V1	S18-V2	S18-V3	S19-V4	S19-V5	S19-V6	S19-V7	S20-V8	S20-V9	S20-V10
SiO_2	74.46	74.22	74.36	69.19	67.41	66.00	67.57	71.91	62.08	74.62
Al_2O_3	11.70	10.02	8.75	13.84	16.09	12.69	10.56	8.10	10.75	10.03
K_2O	4.03	4.81	5.13	4.59	6.71	5.99	6.34	4.92	5.64	4.81
Na_2O	3.94	4.82	5.15	4.78	6.68	6.48	6.77	4.85	7.06	4.82
Total	94.13	93.87	93.39	92.40	96.89	91.16	91.24	89.78	85.53	94.28
Sm	217	857	2,279	238	806	1,389	3,297	4,833	17,674	301
Sm ⁻	138	775	2,180	160	719	1,299	3,183	4,693	17,273	222
Sm ⁺	296	941	2,381	317	893	1,480	3,414	4,978	18,092	380
(Na+K)/Al	0.93	1.31	1.60	0.92	1.14	1.33	1.68	1.65	1.58	0.96
D	1.07	1.55	1.92	1.08	1.42	1.66	2.12	2.33	2.56	1.09

Melt composition determined by microprobe after the experiments. Sm content in ppm. Sm⁺ and Sm⁻ are the upper and lower values, respectively, bracketing the 95% confidence interval. See text and Montel (1986) for analytical details.

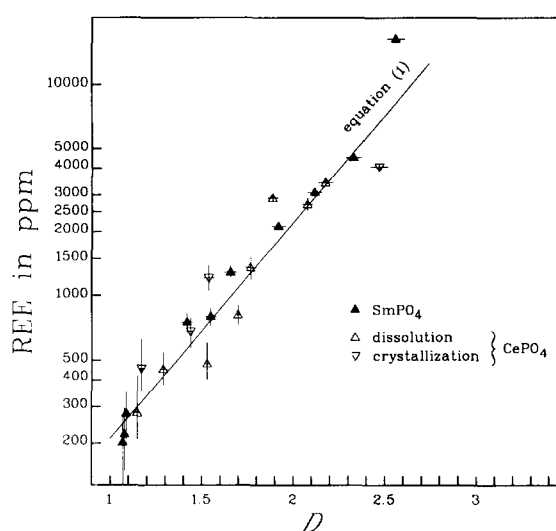


Fig. 3. Effect of melt composition, represented by D , as defined in the text, on monazite solubility, for CePO_4 (Montel, 1986) and SmPO_4 (this study), at 800°C , 2 kbar, water-saturated conditions, in $\text{SiO}_2\text{--Al}_2\text{O}_3\text{--Na}_2\text{O--K}_2\text{O}$ melts.

TABLE 3

Values for W_i/R

La	~260	Nd	177
Ce	~116	Sm	460
Pr	31	Gd	750

These values are for being used in Eq. 3.

Anyway, calculations have shown that even use of an extreme value of 700°C changes the values for W_i only slightly and has almost no effect on the final REE fractionation obtained through Eqs. 2 and 3.

4. Application I: REE thermometry

4.1. Significance and conditions required for REE thermometry

Eq. 1 can be used as a chemical thermometer (REE thermometry). This approach, based on accessory mineral solubility, was initially used successfully by Harrison et al. (1986) and Miller et al. (1988) using zircon, and by Rapp et al. (1987) using monazite. Other examples

of applications, using zircon or monazite, can be found in Brouand et al. (1990), and Scaillet et al. (1990). The temperature indicated by this kind of thermometer is *the temperature at which the accessory mineral last equilibrated with the melt*. This can be:

(1) the temperature of extraction of a granitic magma from its source, in the case that no fractional crystallization occurred between the source region and the emplacement level (Miller et al., 1988);

(2) the temperature of differentiation from a parental magma in case of fractional crystallization (Scaillet et al. 1990);

(3) the temperature of segregation of a leucosome in anatectic migmatites (Brouand et al., 1990);

(4) the temperature prevailing before eruption in the case of the composition of glass in obsidian.

All these temperature are purely magmatic ones, which are difficult or impossible to obtain by other methods. However, thermometers based on accessory mineral solubility operate properly only if: (a) chemical equilibrium prevailed during differentiation; (b) the relevant accessory mineral actually controls the behaviour of the element; and (c) the whole-rock composition represents a frozen liquid.

Chemical equilibrium between melt and crystals [point (a)] is difficult to demonstrate in natural systems. It can be discussed only on a theoretical basis, using measured kinetic parameters, especially diffusion coefficients (Harrison and Watson, 1983, 1984; Rapp and Watson, 1986). Disequilibrium between melt and crystal can result in incomplete dissolution during partial melting (Harrison and Watson, 1983), and "snow plow" effects during fractional crystallization (Green and Watson, 1982; Harrison and Watson, 1984; Bacon, 1989). In both cases, it lowers the REE content of the melt and leads to underestimation of the temperature. In order to reach equilibrium between melt and monazite it is neces-

sary that the crystal "sees" the liquid. This problem was addressed in detail by Watson et al. (1989) who conclude that most of the time accessory minerals remain in contact with the magma.

The presence of monazite during partial melting or fractional crystallization [point (b)] can be verified by a careful examination of the mineralogy of the restites or cumulates (see summary in Barbey, 1991). For example, monazite and zircon have been identified in enclaves from the post-anatectic granites (PAG) of the Velay Dôme (Massif Central, France; Montel et al., 1991). These enclaves have been demonstrated to be restites, indicating that the granites were saturated in REE and Zr from the source. Alternatively, the presence of monazite and zircon in the Macusani tuffs (Pichavant et al., 1988) suggests that their differentiates (the Macusani glasses) were saturated in REE and Zr. As a general rule, it appears that a granite can be considered to be saturated in REE (or Zr) at a level close to its source if the petrographical analysis indicates that monazite (and zircon) formed as an early phase.

One of the major problems with point (c) is the possible presence of restites in the magma. It can be discussed on a petrographical basis, although there is some debate about the criteria to be used to identify proper restites (Chappell et al., 1987; Wall et al., 1987; Clemens, 1988). The presence of inherited monazite is more difficult to demonstrate, but is suspected in some cases (see a review in Barbey, 1992). In view of their similar solubility, density, size and petrographical position in high-grade metamorphic rocks, zircon and monazite should behave in a similar way. However, there is no definitive argument because monazite usually contains no inherited Pb (see, however, Copeland et al., 1988). Detailed petrographical work including microprobe analyses and examination of the internal structures by scanning electron microscope (SEM) can partly solve this problem, as shown

below. Calculations indicate that a *limited* proportion of restite or inherited monazite is not critical for REE thermometry. If 10% of the monazite is inherited, it produces an overestimation of $<10^{\circ}\text{C}$. Similarly, if a magma contains 10% of restite (major minerals), it produces an underestimation of 10°C .

4.2. Applications to various magmatic settings

Examples of application of Eq. 1 for various types of crustal magmatism for which independent P - T estimates are available (including granites, volcanics and migmatites) are presented in Table 4. The results of REE and Zr thermometry, and the P - T estimates by the authors in the original papers are given for each series. For REE thermometry, the values calculated using the calibration of Rapp and Watson (1986) are also reported. The temperatures are calculated for normal types as well as for some highly differentiated members, thought to be formed by extensive differentiation.

Higher Himalayan leucogranites (Manaslu, Vidal et al., 1982; Gangotri, Scaillet et al., 1990) yield relatively low temperatures (below 750°C), in agreement with the estimated P - T conditions of metamorphism for the Tibetan Slab (Pêcher, 1989), and with the generation of granites under fluid-rich conditions (Le Fort et al., 1987; France-Lanord and Le Fort, 1988). An aplite in the Manaslu yields a low temperature ($T < 600^{\circ}\text{C}$), significantly lower than the experimentally determined solidus of Benard et al. (1985) for the granite. Contrarily, the lowest T for the Gangotri pluton ($\sim 630^{\circ}\text{C}$) is compatible with the experimentally determined solidus for this granite (B. Scaillet, pers. commun., 1992).

In cordierite-bearing granites [PAG of the Velay (Massif Central, France; Montel et al., 1992; Montel and Abdelghaffar, 1993), Huelgoat granite (Armorican Massif, France; Georget, 1986) and Carnmenellis granites (southwest England; Charoy, 1986)] the tem-

TABLE 4
Results of REE thermometry for some crustal series

Results of RFL thermometry for some natural series

Series	Type of rock	D	Water content (%)	ΣREE (ppm)	T(REE) (°C)		T(Zr) (°C)	Ether estimate (°C)	Reference(s)	
					this study	R & W				
Manaslu, Higher Himalaya Gangotri, Higher Himalaya Macusani, Peru	Ms-Bt granites, aplite	1.03-1.01	6	35-70	678-719	678-718	661-623	700	Vidal et al. (1982)	
	Tur-Bt-Ms granites	1.07	6	6.2	564	579	594		Pécher (1989)	
			9.98-1.06	6	17-58	632-717	635-710	636-705		Scaillet (1990)
										Scaillet et al. (1990)
Post-anatectic granites, Velay, Massif Central, France	Bt-Ms tuffs, And obsidian	0.89-1.06	3	65-120	752-809	752-809	718-764	800	Pichavant et al. (1988)	
	Crd-Bt granites, Ms-Bt granite, Ms-Toz granite	0.99-1.00	5.5	6.7-11	607-581	608-584	665-674	650	Pichavant et al. (1987a)	
			0.95-1.16	4	175-290	801-849	829-789	797-872	800-850	Montel et al. (1992)
			1.00	6	81	739	731	739		
Carmenellis, Cornubian batholith, SW England	Ms-Bt-Ms granites, Ms-Toz-Ms granites, Ms-Toz-Tur aplite	1.00	6	26	658	658	654			
		0.97-1.00	4	94-158	768-813	742-800	728-769	800	Charoy (1986)	
		0.92	6	6.3	584	587	631			
Huelgoat, Brittany, France	Crd-Bt-Ms granites, aprites and Ms granite	0.86-1.22	4	43-143	715-802	693-757	700-793	700-800	Georget (1986)	
	Bt-Crd-Grt-Opx granites, leucogranite	0.86-0.90	6	21-35	679-707	646-696	654-693			
			0.73-1.35	4	294-105	786-892	771-830	770-875	780-900	Charoy et al. (1990)
			1.12	6	31	650	668	633		
Migmatite, Beaune Valley, Ardèche, France	in situ leucosome, injected dyke69	0.81-0.03	6	126-144	769-823	757-764		690-730	Weber and Barbey (1986)	
		1.02	6	21	669	672				
Migmatite, Tibetan Slab	in situ and injected leucosome	0.87-1.16	6	32-54	658-698	757-764	671-683	710	Brouand et al. (1990)	

For each series results are presented for the main types (first line) and for some highly differentiated terms (second line). D is defined in the text. H₂O is the water content given by the author in the original papers. If unknown, 4% was taken for the main types and 6% for the differentiated rocks. Σ REE: sum of REE contents in ppm, including calculated Pr, but excluding Eu. T(REE), this study: temperature calculated using Eq. 1. T(REE), R & W: temperature calculated using the calibration on Rapp and Watson (1986). T(Zr): temperature given by zircon thermometry. "Other estimate" is the value given by the author, in the original papers, for the generation of the granitic magmas. Abbreviations from Kretz (1983).

peratures reach higher values; the PAG yields high T (800–850°C) compatible with direct thermometry performed on the enclaves (Montel et al., 1992). Other types give lower T , in the range 700–820°C, compatible with generation by fluid-absent melting of a pelitic protolith (Green, 1976; Clemens and Vielzeuf, 1987; Vielzeuf and Holloway, 1988; Patiño-Douce and Johnston, 1991). The low temperature found for the very differentiated samples are in agreement with the temperature of the granitic minima (Tuttle and Bowen, 1958; Luth et al., 1964; and summaries in Johannes, 1985, and Johannes and Holtz, 1990). However, for the Carnmenellis pluton, the temperature for the aplites (580°C) is too low, even if the effect of fluorine on the granitic minima is taken into account (Pichavant and Manning, 1984).

The Darongshan batholith (China) consists of Opx–Grt–Crd granites (Charoy et al., 1990), very similar to the true Australian S-type (White and Chapell, 1977). Temperatures are very high (800–900°C), compatible with generation of those granitoids by a high degree of melting of a metasedimentary protolith. However, results for this suite must be taken with caution because it is suspected to contain a large amount of restites, and because most compositions are rich in FeO, MgO and CaO.

The two examples of migmatites display contrasting results. The in situ migmatites from the Beaume Valley (Ardèche, France; Weber and Barbey, 1986) yield a high temperature (780–840°C), incompatible with the results of conventional thermometry, and with the apparently low degree of melting of these rocks. This suggests that the leucosomes do not represent crystal free-liquids saturated in monazite, but contain a significant proportion of undissolved monazite crystals. On the contrary, the temperature found for an injected dyke (681°C), can be considered to be the temperature of generation of this melt. The leucosomes of the Tibetan Slab studied by

Brouand et al. (1990) yield temperatures nearly identical to those obtained by Zr thermometry, consistent with partial melting under water-saturated conditions.

4.3. General considerations

A comparison of temperatures calculated by REE (Eq. 1) and Zr thermometry is shown in Fig. 4. The agreement between the two thermometers is good, with an average difference of 25°C. The maximum difference is for $T < 700^\circ\text{C}$. It is possible that additional constituents not taken into account by available equations (in particular volatiles) disturb the behaviour of one or both thermometers, as suggested by Keppler (1992). For the series investigated, the two calibrations of REE thermometry yield almost identical results. As discussed in Section 2.3, both formulations yield

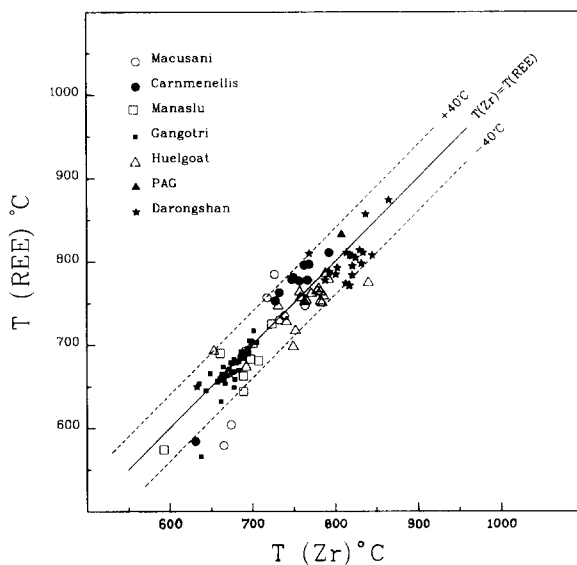


Fig. 4. Comparison between the temperature obtained by Zr thermometry (Watson and Harrison, 1983) and REE thermometry, Eq. 1, for the series presented in Table 4.

similar results in the case of peraluminous or metaluminous natural composition with a "normal" water content.

Three arguments confirm the reliability of REE and Zr thermometry:

(1) There is a very good agreement between the two thermometers, which have been calibrated independently. This was expected for low-Ca granites, but it is more surprising for less differentiated facies because Eq. 1 is normally valid only for low-Ca melts. Agreement between the REE and Zr thermometry also suggests that monazite and zircon are extracted from granitic melts during the same magmatic events.

(2) The highest temperature found for each granitic series is compatible with the temperature required for partial melting of metasediments (Green, 1976; Lebreton and Thomson, 1988; Conrad et al., 1988; Vielzeuf and Holloway, 1988; Johannes and Holtz, 1990; Montel and Vielzeuf, 1991; Patiño-Douce and Johnston, 1991; Vielzeuf and Montel, 1991). Moreover, the difference from one series to another is compatible with the difference in the suspected generation mechanism: fluid-present melting for the Higher Himalayan leucogranites; fluid-absent melting of pelites for Carnmenellis. PAG and Huelgoat granites; fluid-absent melting with high degree of melting for S-types.

(3) The lowest T found in each series is usually close to those of known minima in granitic systems.

High temperatures are systematically given by facies enriched in Fe, Mg and Ca, whereas low temperatures are encountered in Fe-, Mg- and Ca-poor, volatile-rich granites, aplites or pegmatites. In intermediate facies, the temperature decreases as differentiation proceeds. It can be reasonably assumed that, in a given series, the highest T represents the temperature of genesis in the source, the lowest T represents the last magmatic activity with expulsion of residual melts, and intermediate T represents several stages of differentiation.

5. Application II: monazite as a petrogenetic indicator

5.1. Monazite: textures, composition, structural formula

Using Eqs. 1–3, it is possible to calculate the REE pattern of a liquid (from La to Gd) in equilibrium with a monazite of known composition. It is possible also to discuss genetic relationships between magmas by comparing the composition of the monazites of the presumed parental rock and the REE pattern of the presumed daughter rock. This can be used to identify the source rock of a granite, or the parental magma of a residual melt. It can also be used to discuss the degree of equilibrium between leucosomes and melanosome in migmatites.

The first step is to analyse monazite. Because it has variable composition and internal zonation, an electron microprobe should be used. Best results have been obtained, in Clermont Ferrand, using the following conditions: accelerating voltage: 25 kV, sample current: 30 nA, counting time: 30 s on peaks, 10 s on background; K_{α} line used for P, Si and Ca, L_{α} for La, Ce and Y, L_{β} for Pr, Nd, Sm and Gd, M_{γ} for U and Th. This choice guarantees minimal interferences for X-ray lines (Roedder, 1985). Standards are synthetic $CePO_4$, $LaPO_4$, $NdPO_4$ and YPO_4 for Ce, La, Nd and Y respectively (Montel et al., 1989); glasses for Sm, Gd and Pr (Drake and Weill, 1972); ThO_2 ; UO_2 ; albite for Si; apatite for Ca; and YPO_4 for P.

All analyses have been carried out on thin sections. Before proceeding, all the grains in each thin section which optically looked like zircon were studied using an SEM equipped with an EDS (energy-dispersive spectrometric) system. This method unambiguously distinguished monazite from zircon and xenotime. Using backscattered electron imaging, it was also possible to reveal the internal structures of monazite crystals, this being used as a guide during microprobe analysis. Analytical

TABLE 5

Microprobe analyses and structural formula of monazites from the Tibetan Slab

	M114						JANU III				
	5	7	8	9	10	11	1	2	3	4	5
ThO ₂	5.15	5.23	5.66	5.57	6.62	7.36	6.94	8.39	8.33	7.42	6.67
P ₂ O ₅	30.71	30.47	28.94	30.74	30.92	30.81	30.67	29.94	28.94	30.51	29.05
Y ₂ O ₃	0.21	0.20	0.40	0.32	0.15	0.24	2.68	2.45	1.30	3.03	1.30
CaO	1.47	1.54	1.44	1.50	1.66	1.85	1.66	1.25	0.61	1.74	0.56
UO ₂	0.83	0.81	0.81	0.94	0.96	1.07	1.54	0.51	0.42	1.32	0.42
Gd ₂ O ₃	1.27	1.39	1.47	1.42	1.29	1.30	1.28	1.19	0.77	1.46	0.68
Sm ₂ O ₃	1.89	1.92	2.17	2.16	2.26	1.78	1.66	1.54	1.16	1.74	1.18
Nd ₂ O ₃	11.63	11.88	12.40	12.32	12.11	11.01	9.64	9.85	10.24	10.00	10.23
Pr ₂ O ₃	2.98	3.45	3.19	2.92	3.03	2.72	2.77	2.91	2.84	2.65	2.84
Ce ₂ O ₃	28.53	28.21	27.65	27.72	26.95	25.91	27.54	27.89	29.90	27.05	29.97
La ₂ O ₃	15.36	14.87	13.98	13.66	13.73	14.19	14.13	13.86	15.99	13.82	16.02
SiO ₂	0.36	0.34	0.48	0.41	0.44	0.44	0.27	0.75	1.53	0.36	1.20
Total	100.39	100.31	98.59	99.68	100.12	98.68	100.78	100.53	102.03	101.10	100.12
Si	0.014	0.013	0.019	0.016	0.015	0.016	0.010	0.029	0.059	0.014	0.047
P	0.986	0.987	0.981	0.984	0.985	0.984	0.990	0.971	0.941	0.986	0.953
La	0.215	0.210	0.206	0.191	0.205	0.203	0.199	0.196	0.227	0.195	0.229
Ce	0.396	0.395	0.405	0.384	0.395	0.395	0.384	0.391	0.421	0.378	0.425
Pr	0.041	0.048	0.047	0.040	0.044	0.045	0.038	0.041	0.040	0.037	0.040
Nd	0.158	0.162	0.177	0.166	0.166	0.168	0.131	0.135	0.140	0.136	0.142
Sm	0.025	0.025	0.030	0.028	0.027	0.028	0.022	0.020	0.015	0.023	0.016
Gd	0.016	0.018	0.020	0.018	0.018	0.019	0.017	0.016	0.010	0.019	0.009
Y	0.004	0.004	0.009	0.006	0.006	0.006	0.054	0.050	0.027	0.062	0.027
U	0.007	0.007	0.007	0.008	0.007	0.007	0.013	0.004	0.004	0.011	0.004
Th	0.044	0.046	0.052	0.048	0.047	0.048	0.060	0.073	0.073	0.064	0.059
Ca	0.060	0.063	0.062	0.061	0.061	0.062	0.068	0.051	0.025	0.071	0.023
(La/Gd) _N	9.62	8.51	7.57	7.65	8.30	8.00	8.78	9.27	16.52	7.53	18.74

Structural formula calculated assuming Si + P = 1. (La/Gd)_N: La/Nd normalized to chondrites. Numbers refers to the data points in Fig. 5a and b.

results indicate that the internal structure detectable by SEM corresponds to chemical variations for all the elements (Tables 5 and 6; Fig. 5).

Since Si is the only atom that substitutes for P in monazite (Deer et al, 1962; Clark, 1984) the structural formula is calculated assuming Si + P = 1 in the tetrahedral site, all other elements being assigned to the seven-coordinated site (Smyth and Bish, 1988). This method has the advantage of giving a correct formula even for partial analyses. The mole fraction for each REE-phosphate and for YPO₄ is given directly by the number of REE atoms in the formula unit. The mole fractions of the Th and U end-

members are calculated assuming that U and Th are equally distributed between huttonite-like (ThSiO₄, USiO₄) and brabantite-like (CaTh(PO₄)₂, CaU(PO₄)₂) end-members.

5.2. Application to the Manaslu granite

Three thin sections have been studied in detail: one from the Manaslu granite, and two from the Tibetan Slab. In the studied samples, only one-half of the grains that looked like zircon are actually zircon, the remaining 50% were revealed to be monazite ($\frac{4}{5}$) and xenotime ($\frac{1}{5}$). They occur pervasively through the

TABLE 6

Microprobe analyses and structural formula of monazites from the Manaslu granite

	Core 1	Medium 2	Rim 3	Rim 4	Nucleus 7	Nucleus 9	Core 10	Middle 11	Rim 12	Rim 13	Rim 14
ThO ₂	6.25	8.49	5.64	6.64	7.43	8.55	3.42	5.51	6.67	7.24	6.24
P ₂ O ₅	30.34	30.18	30.52	30.36	30.39	30.41	29.74	30.25	29.88	29.96	30.16
Y ₂ O ₃	4.57	3.80	3.18	2.87	2.65	2.44	1.69	2.58	2.99	3.90	2.76
CaO	1.39	1.63	1.49	1.65	2.03	2.03	1.94	1.35	1.26	1.37	1.70
UO ₂	0.35	0.31	1.84	1.40	2.38	1.36	0.02	0.03	0.21	0.13	1.76
Gd ₂ O ₃	2.10	2.09	2.01	1.99	2.18	1.96	1.15	1.68	2.09	2.25	1.89
Sm ₂ O ₃	2.45	2.54	2.52	2.63	2.72	2.69	1.45	2.00	2.50	2.44	2.63
Nd ₂ O ₃	10.92	11.86	11.34	11.34	10.75	11.21	11.38	11.56	11.95	12.02	11.46
Pr ₂ O ₃	2.70	2.97	2.98	2.97	2.86	3.12	3.23	2.96	2.92	2.98	2.89
Ce ₂ O ₃	25.36	24.88	26.48	25.81	25.05	24.94	29.68	27.69	26.20	25.70	25.67
La ₂ O ₃	12.50	11.27	12.51	12.26	11.62	11.76	15.86	14.21	12.28	11.35	11.94
SiO ₂	0.25	0.31	0.24	0.41	0.29	0.33	0.21	0.27	0.47	0.39	1.90
Total	99.18	100.33	100.75	100.33	100.35	100.80	99.77	100.09	99.42	99.73	101.00
Si	0.010	0.012	0.009	0.016	0.011	0.013	0.008	0.010	0.018	0.015	0.069
P	0.990	0.988	0.991	0.984	0.989	0.987	0.992	0.990	0.982	0.985	0.931
La	0.178	0.161	0.177	0.173	0.165	0.166	0.230	0.203	0.176	0.163	0.161
Ce	0.358	0.352	0.372	0.362	0.352	0.350	0.428	0.392	0.372	0.365	0.343
Pr	0.038	0.042	0.042	0.041	0.040	0.044	0.046	0.042	0.041	0.042	0.038
Nd	0.150	0.164	0.155	0.155	0.148	0.154	0.160	0.160	0.166	0.167	0.149
Sm	0.033	0.034	0.033	0.035	0.036	0.036	0.020	0.027	0.033	0.033	0.033
Gd	0.028	0.028	0.026	0.026	0.029	0.026	0.015	0.022	0.028	0.030	0.024
Y	0.094	0.078	0.065	0.058	0.054	0.050	0.035	0.053	0.062	0.081	0.054
U	0.003	0.003	0.016	0.012	0.020	0.012	0.000	0.000	0.002	0.001	0.014
Th	0.055	0.075	0.049	0.058	0.065	0.075	0.031	0.048	0.059	0.064	0.052
Ca	0.057	0.068	0.061	0.068	0.084	0.083	0.082	0.056	0.052	0.057	0.066
(La/Gd) _N	4.74	4.29	4.95	4.90	4.24	4.77	10.97	6.73	4.67	4.01	5.03

Structural formula calculated assuming Si + P = 1. (La/Gd)_N: La/Nd normalized to chondrites. Numbers refers to the data points in Fig. 5c and d.

rock, as inclusions in biotite, and as interstitial crystals at the grain boundary between two minerals (thus confirming the results of Watson et al., 1989). Monazites are larger (50–200 μm) in the Tibetan Slab thin sections than in the Manaslu granite (20–80 μm). The internal structure of the monazites from the Tibetan Slab (Fig. 5a and b) shows irregularly-shaped, but apparently homogeneous, juxtaposed areas. This will be referred to as “foie-gras zoning” (because of its similarity with the internal structure of “foie-grass”). In contrast, the internal structures of the monazites of the Manaslu granite are regular and euhedral in shape (Fig. 5c and d). In one case a small nucleus

has been observed (Fig. 5d). The absence of large monazites with “foie-gras zoning” in the granite suggests that there is no *extensive* inheritance of monazite from the Slab, even if the small nucleus in the monazite depicted in Fig. 5d can be interpreted as an inherited core.

In the Tibetan Slab, the analyzed monazites (Table 5) have relatively constant compositions, characterized by high Th and U contents and a high (La/Gd)_N ratio (La/Gd normalized to chondrite). They are also characterized by a high degree of brabantitic substitution (high Ca content). By comparison, the monazites from the Manaslu granite display important internal variations. The discontinuity from

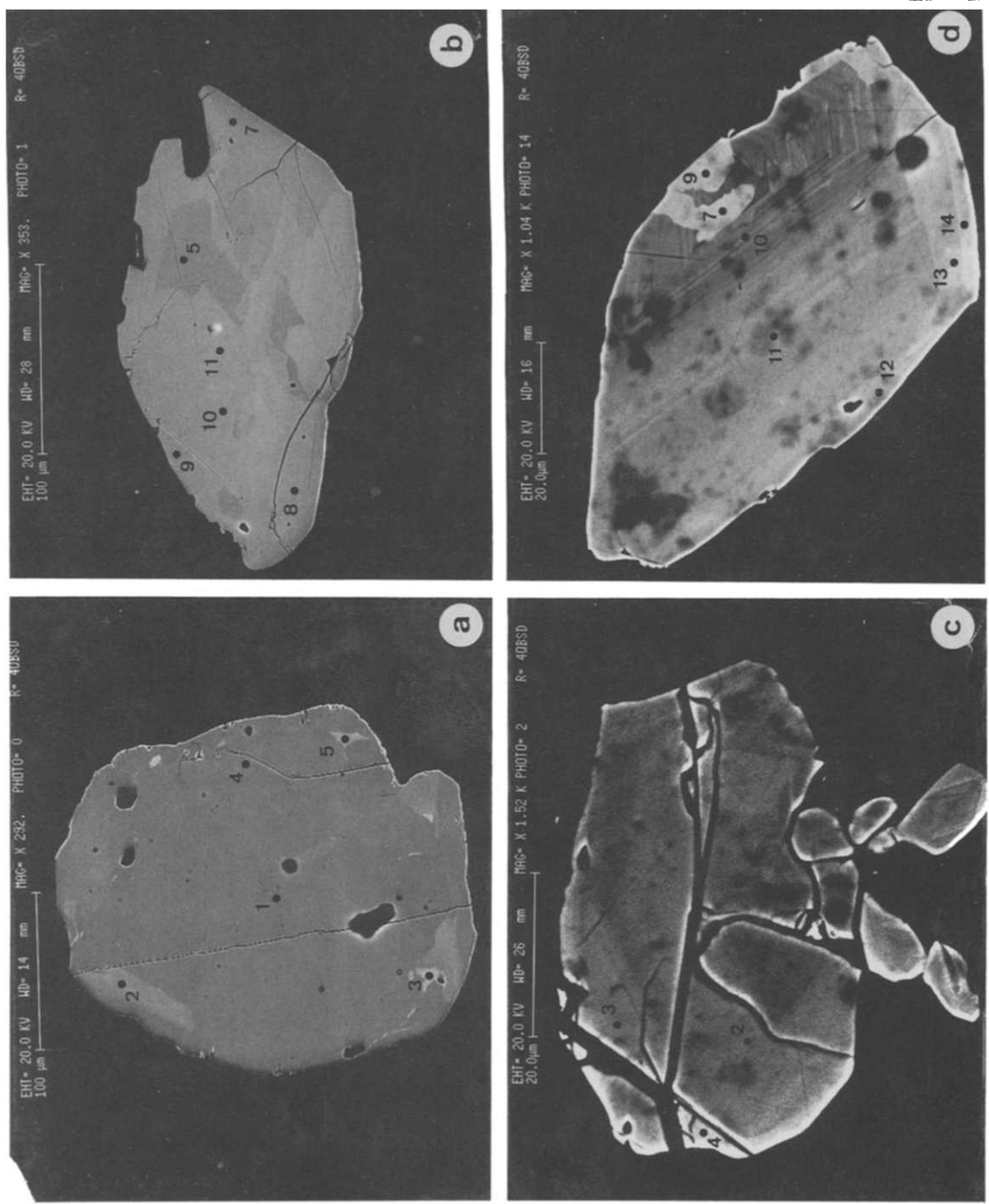


Fig. 5. Backscattered electron images of monazite. *Numbers* refer to analyses in Tables 5 and 6.
a. Sample JANU III, Tibetan Slab. This crystal is situated at the grain boundary between two biotites.
b. Sample M114, Tibetan Slab. This crystal is included in a kyanite, and shows typical "foie-gras zoning".
c. and d. Monazite from the Manaslu granite, sample U900. Crystal (c) is situated at the grain boundary between a biotite (down) and a plagioclase (up). Crystal (d) is situated at the grain boundary between a biotite (up) and a quartz (down). Both display euhedral zoning, with a nucleus in monazite (d).

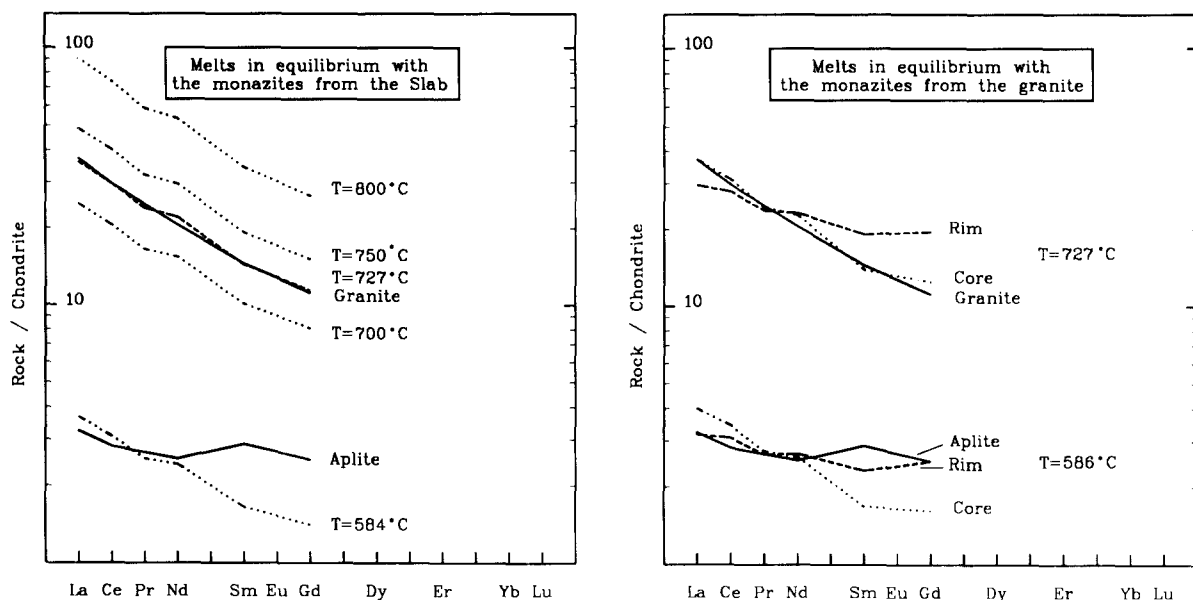


Fig. 6. Whole-rock REE patterns of the Manaslu granite and aplite (Vidal et al., 1982; samples D22 and N67), respectively, and calculated REE patterns for melts in equilibrium with the monazites of the Tibetan Slab (sample M114, composition 7 in Table 5) and with the monazite of the granite [monazite (d), compositions 10 for the core and 13 for the rim, Table 6]. The temperatures are those used in Eqs. 1–3.

the nucleus to the core in Fig. 5d corresponds to an abrupt change in chemical composition, whereas the composition changes progressively from the core to the rim, with increasing Th and U content and decreasing La/Sm ratio.

The composition of the melt in equilibrium with monazites from the slab has been calculated at various T , with 6% H_2O . Results are compared to the REE patterns of the Manaslu granites (Vidal et al., 1982) Fig. 6a. The REE abundance depends on the temperature, but the shape of the pattern depends mainly on the composition of the monazite. The calculated REE distribution of the melt is nearly identical to the REE pattern of the Manaslu granite for $T=727^{\circ}\text{C}$, a temperature identical to that found by other methods (Zr thermometry, Table 4, and Pêcher, 1989). Consequently, the Manaslu granites can be considered to represent a melt in equilibrium with the monazites of the slab. This is consistent with the fact that the Manaslu granites represent a product of partial melting of the Tibetan Slab under monazite-saturated conditions. Contrarily, it is im-

possible to reproduce the pattern of the aplite at any T (Fig. 6a). This suggests that the aplite is not a direct product of partial melting of the Tibetan Slab or requires a protolith containing monazites of a totally different composition, or no monazite at all.

The REE distribution of melts in equilibrium with a monazite from the Manaslu granite (monazite shown in Fig. 5d) has been calculated at various T . Both core and rim compositions were used (Table 6). The calculated melt REE patterns are compared to the REE patterns of the granites and aplite in Fig. 6b. It is possible to fit the REE pattern of the granite using the composition of the core for $T=727^{\circ}\text{C}$. Contrarily, it is impossible to equilibrate the rim of the monazite with the REE pattern of the granites (Fig. 6b). Conversely, the REE pattern of the aplite can be in equilibrium with the rim of the monazite at $T=586^{\circ}\text{C}$, but not with the core at any temperature. This indicates that the core of this monazite corresponds to the first crystal formed in the melt (in this case growing on a

partially undissolved nucleus), and that the aplite represents a residual melt in equilibrium with the lastly formed monazite.

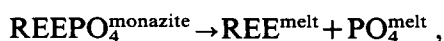
6. Discussion

6.1. Limitations and possible extensions of the model

The model presented here, although giving encouraging results, should probably be used only for peraluminous or metaluminous, Ca-poor series at low temperatures. Very high temperatures obtained by REE thermometry (above 900°C, or even 850°C) must be considered with caution, because it is very likely that monazite is totally dissolved at such high temperature. In Ca-rich systems, allanite can be present (Lee and Bastron, 1967; Cuney and Friedrich, 1987) and monazite may not buffer the REE. Dubious results may be obtained in volatile-rich or mafic systems, because the effects of high volatile content and high Fe–Mg content have not been investigated experimentally. The effects of melt composition on REE fractionation is unknown, and it is probably unreliable to use Eqs. 2 and 3 for liquids with D outside of the interval 0.9–1.1. In modelling the solubility of Th and U end-members, additional equations would certainly be necessary, but this would give some insight into the behaviour of U and Th in the continental crust. It must be emphasized that detailed petrographical observation should be used as a “garde-fou” for this kind of thermometry.

6.2. The role of phosphorus in monazite solubility

The mechanism of dissolution of monazite in felsic melt is not known precisely. The dissolution of monazite has been proposed to operate by the reactions (Rapp and Watson, 1986; Montel, 1986):



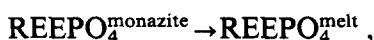
$$K = \frac{a_{\text{REE}}^{\text{melt}} \cdot a_{\text{PO}_4}^{\text{melt}}}{a_{\text{REEPO}_4}^{\text{monazite}}}$$



$$K = \frac{a_{\text{REE}_2\text{O}_3}^{\text{melt}} \cdot a_{\text{P}_2\text{O}_5}^{\text{melt}}}{\left(a_{\text{REEPO}_4}^{\text{monazite}}\right)^2}$$

where a_j^i = activity of component j in phase i .

In both cases the solubility of monazite should depend on the activity of phosphorus in the melt, as proposed by Montel (1986). The real effect of phosphorus cannot be discussed from experimental data because all the available data were obtained with a molar ratio REE/P equal to 1. This is not the case for natural rocks, so P_2O_5 should be taken into account. The activity of phosphorus in granitic magmas is very difficult to estimate because it can form complexes in peraluminous melts (Pichavant et al., 1992). For such melts, it is not possible to estimate the activity of P_2O_5 from its content in the melt, even for a liquid saturated in apatite. Two extreme cases, which yield two equations similar to Eq. 1 but including phosphorus, were being investigated: in one case the activity of P_2O_5 was assumed to be equal to the concentration; in the other case, the activity of phosphorus was assumed to be buffered by apatite, according to the model of Harrison and Watson (1984). Application of those equations to the natural occurrences presented in Table 4 always produces unrealistically high temperatures (>900°C). This means that monazite should be almost totally insoluble in felsic melts, which disagrees with petrologic observations. In particular, both equations predict that monazite in the Manaslu granite should be totally inherited from the Tibetan Slab, which is not compatible with the data presented above. This suggests that monazite solubility is independent of phosphorus activity, which is possible if the mechanism of dissolution of monazite is something like:



$$K = \frac{a_{\text{REEPO}_4}^{\text{melt}}}{a_{\text{REEPO}_4}^{\text{monazite}}}$$

In this model REE and phosphate are associated in a monazite-like complex, similar to the berlinite-like units proposed by Pichavant et al. (1992), which also involves phosphorus. The presence of such units has been proposed by Ryerson and Hess (1978) to explain the preferential partitioning of REE toward P_2O_5 -rich melts in coexisting immiscible silicate liquids. It is also supported by the experimental results of Rapp and Watson (1986), who found that REE and phosphorus diffuse together during partial dissolution of monazite. However, further experimental and spectroscopical work would be necessary to demonstrate the existence of such units in silicate melts.

7. Conclusions

The combination of experimental data and natural compositions has led to a model for equilibrium between melt and monazite. This permits one to calculate not only the amount of REE dissolved in the melt but also the REE distribution, from La to Gd. This set of equations can be used in two ways. The first is REE thermometry which, when used concurrently with Zr thermometry, is a powerful tool to study magmatic processes in peraluminous felsic suites. The second requires more specific work (including microprobe analyses of monazites) but allows one to discuss the generation and the differentiation of crustal melts, and, the genetic relationships between magmas.

Because of its importance in geochronology, monazite has received considerable attention. The results presented here suggest that this mineral is important not only because it is rich in some commonly used isotopes, but also as an indicator of magmatic temperatures and petrogenetic relationships in granites.

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